

[DC] Context-Aware Inference and Adaptation in Augmented Reality

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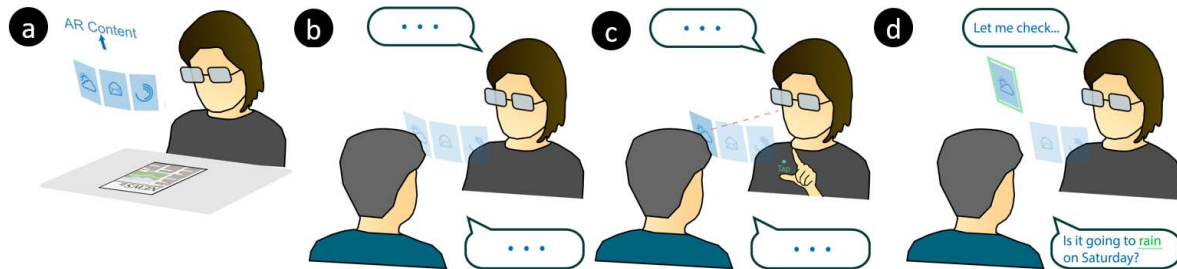


Figure 1: Proper knowledge of the AR user's context allows for design of highly efficient yet less intrusive interfaces. For example, (a) when not occluding any object of importance in the real world, always-available virtual content improves information access efficiency. However, in a social context where issues such as occlusion, distraction, and visual clutter are predominant, it is preferred to (b) keep the virtual content transparent when not needed and (c) give the user control to manually summon the virtual content when desired. (d) Context-awareness also enables automatic access to the relevant virtual content that supports the user's primary task, in the least intrusive manner.

ABSTRACT

Augmented Reality (AR) offers the potential for easy and efficient information access, reinforcing the wide belief that AR Glasses are the next-generation of personal computing devices. However, to realize this all-day AR vision, the AR interface must be able to address the challenges that constant and pervasive presence of virtual content can cause for the user. The optimal interface, that is the most efficient yet least intrusive, in one context may be the worst interface for another context. Throughout the day, as the user switches context, an optimal all-day interface must adapt its virtual content display and interactions as well. This work aims to propose a research agenda to design and validate different adaptation techniques and context-aware AR interfaces and introduce a framework for the design of such intelligent interfaces.

Index Terms: Human-centered computing—Mixed / augmented reality; Human-centered computing—Interaction techniques; Human-centered computing—Empirical Studies in HCI

1 INTRODUCTION

Augmented Reality (AR) glasses have long been envisioned to be the next-generation of personal computing devices, becoming “much like telephone and PC ... displaying information we expect to see both at work and at play” [3]. The recent hardware developments promise a near future in which powerful, lightweight AR glasses can be worn all-day and fulfill ubiquitous information needs of users [7] through pervasive, universal, and omnipresent access to digital content, enabling continuous access to personal apps in everyday scenarios and tasks [4]. This will be a future in which people can check the weather forecast and their personal messages as easily and as quickly as a glance without any manual interaction. However, hardware development is not the only limiting factor to the realization of this envisioned future.

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The AR interface itself must also handle AR challenges such as avoiding distraction, visual clutter and occlusion of important real-world objects [1], identifying the desired virtual content and its appropriate position, placement, accessibility, size, transparency, and level of detail (LoD), and determining the proper interaction technique. As Grubert suggests, the pervasive AR future relies on the design of an automatic AR system that “can adapt to the current, changing requirements and constraints of the user's context and thus allows for a continuous usage.” [4].

Recent research has shed light on both automatic and non-automatic, user-triggered adaptive interfaces in AR. Lindlbauer et al. proposed an online context-aware method to automatically adapt the position and LoD of virtual information [6]. Lages and Bowman proposed an interface to adapt the placement and position of digital content represented as 2D windows in 3D space through manual interactions on a handheld controller [5]. Very few studies have, however, researched automatic adaptation to not only “direct” contextual changes, but also “indirect” ones, which are initiated by the actions of others or environmental changes, and are often uncontrollable by the users themselves. The interfaces introduced in the previous work are also limited to either focusing on a limited number of AR challenges, or a limited set of contexts. The goals of this research are to: **RG1.** explore specific adaptation mechanisms in specific contexts, **RG2.** prototype and validate the overall benefits of context-aware AR interfaces in a limited set of contexts, and **RG3.** propose a holistic framework for the design of such intelligent AR interfaces. The proposed framework aims to identify a comprehensive description of everyday user contexts, infer a set of desired adaptations from it, and specify ways to adapt the AR interface based on the inferences.

To achieve **RG1**, in our previous work we proposed multiple adaptation technique design dimensions and evaluated different techniques designed based on them in a real-world monitoring context [1]. We have also prototyped a context-aware AR interface in two common usage contexts and conducted a user study to validate its benefits in order to pursue **RG2**. In our future work, to further pursue this research goal we plan to prototype and validate the benefits of context-aware interfaces in other contexts as well. Currently, we are focused on **RG3** and identifying a comprehensive description

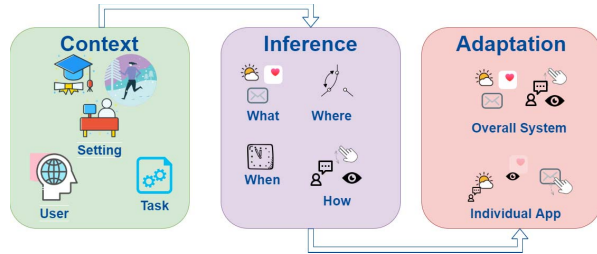


Figure 2: Context-Aware inference and adaptation framework.

of everyday user contexts from which a more complete set of desired adaptations can be inferred.

2 ADAPTATION TECHNIQUE DESIGN

In our first study [1] we introduced three different design dimensions for adaptation techniques: content prioritization, adaptation mechanism, and level of automation. Using these design dimensions we proposed eight adaptation techniques and evaluated information access in a specific context that required the user to monitor the occurrence of a specific event in the real-world. To evaluate these techniques in a controlled manner, we ran an AR simulation experiment in virtual reality.

The results showed that techniques that prioritized the real-world and the ones that automatically detected the real-world object of importance and adapted, resulted in better task performance, and were preferred, particularly when translucency was used as the adaptation mechanism. Despite the ease of information access that prioritizing the virtual content offered, these techniques were less desirable in this context, which confirms the significance of context-awareness. This was mainly due to the primary task's reliance on the real world and the user's interest in minimizing visual clutter and information overload.

3 BENEFITS OF CONTEXT-AWARE AR

We designed a context-aware AR interface to intelligently adapt to both direct and indirect changes in the user's context [2]. The interface was designed for both a static solo context and a dynamic social context. In the solo context, the interface prioritizes the virtual content and places it at the user's eye-level (Fig. 1, a). When the context-aware interface detects the user's context switch to a social conversation, it prioritizes the real world (Fig. 1, b) and based on the conversation's content, automatically makes the related virtual content available in a non-occluding manner (Fig. 1, d). We evaluated information access efficiency, intrusiveness to the primary task and user experience using context-aware AR compared to current mobile phones and non-adaptive Glanceable AR interfaces.

In the solo context, compared to a mobile phone interface, the context-aware AR interface was preferred, easier, and significantly faster; it improved the user experience; and it allowed the user to better focus on their primary task. *In the social context*, context-aware AR was faster for responding to information needs triggered by the conversation; it was preferred and perceived as the easiest for resuming conversation after information access; and it improved the user's awareness of the other person's facial expressions. Our results show the advantages of context-aware AR.

4 CONTEXT-AWARE INTERFACE DESIGN FRAMEWORK

We are currently in the process of developing a holistic framework for the design of context-aware AR interfaces (Fig. 2). We are working to enhance a previous taxonomy [4] and propose a comprehensive description for everyday user contexts that is aligned with our goals. We believe that user context can be described by three main components: (1) users and their personal characteristics, (2)

user's tasks, and (3) setting and environmental features. Having proper knowledge of these components, an intelligent AR interface can infer users' wants and needs, the desired virtual content and interaction techniques, and the appropriate time, place and visual features for displaying that content. Depending on the context, the intelligent interface may make general adaptations to the whole system, or to individual apps.

5 FUTURE WORK

With a reliable and detailed framework, we will choose additional user contexts that have not been studied. We will then upgrade our current context-aware prototype to detect and adapt to these contexts. For example, when cooking, the user needs access to the always-available recipe. Here, since the user's environment is static, to insure consistency and prevent physical hazard to the user, the recipe can attach to a wall above the stove where it is the easiest to glance at, yet less dynamic and obtrusive. In this context, since the primary task of cooking requires high attention, only the the recipe must be presented, displaying the next step in a low LoD. However, when walking down a busy street, to ensure the user's safety and awareness of their ever-changing environment, visibility of the real-world must be prioritized over the ease and efficiency of information access. In this context, manually summoned virtual content that follows the user may be preferred to always-available virtual content that is attached to a separate real-world object. Once we design a reliable context-aware prototype that adapts to the selected contexts, we will conduct one or more user studies to validate the benefits of the proposed context-aware interface.

6 CONCLUSION

The chief contributions of this dissertation will be: (1) prototyping various adaptation techniques and context-aware AR interfaces for multiple contexts, (2) evaluation and validation of the benefits of context-aware AR interfaces, and (3) a holistic framework for context-aware inference and adaptation in AR.

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